

UPI Impulse Trap: Behavioral Intelligence and Financial Regret Analysis Among Indian College Students

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Abstract

Unified Payments Interface (UPI) has transformed digital spending behavior among Indian college students by enabling frictionless transactions and near-instant purchasing. While this convenience improves accessibility, it may also increase impulsive spending and post-purchase financial regret. This study investigates the behavioral and psychological drivers of UPI-based impulse spending using primary survey data collected from 105 students across four Indian institutions.

The project combines Exploratory Data Analysis (EDA), Natural Language Processing (NLP), K-Means clustering, and Random Forest classification to analyze spending behavior, emotional triggers, regret patterns, and financial risk indicators. Results reveal that evening and late-night hours dominate impulsive purchases, food delivery is the most regretted category, and stress-related spending is more influential than social pressure or promotional triggers. K-Means clustering identifies three behavioral spending personas, while Random Forest feature importance analysis highlights impulse composite score and transaction frequency as the strongest predictors of financial regret.

An interactive Streamlit dashboard was developed to visualize behavioral patterns, spending personas, and financial risk predictions. The study demonstrates how behavioral analytics and machine learning can be applied to understand digital payment psychology among students.

Keywords

UPI, impulse buying, financial regret, college students, K-Means clustering, Random Forest, NLP, behavioral analytics, India

1 Introduction

India crossed 500 million Unified Payments Interface (UPI) users in 2024, with college students among the most active adopters. UPI enables near-instant digital payments through mobile applications such as Google Pay, PhonePe, and Paytm, removing traditional friction associated with physical cash or card transactions. Behavioral economists describe the “pain of paying” phenomenon [2]: the more abstract and invisible a payment method, the lower the psychological resistance to spending. UPI represents the most frictionless payment mechanism available in the Indian context, with no physical card, no visible cash depletion, and no PIN delay in many configurations.

This frictionlessness has a behavioral consequence. College students operating on limited monthly budgets, often funded entirely

by parents, are exposed to an environment where digital spending requires minimal deliberation. Food delivery platforms, quick-commerce applications, and online marketplaces deliver gratification within minutes of a UPI transaction. The result is a pattern of unplanned purchases followed by post-purchase regret that is understudied in the Indian college context.

This project makes the following contributions:

- A primary dataset of 105 college students across four Indian institutions covering spending behavior, impulse triggers, and financial regret, collected via a structured Google Form survey.
- Exploratory analysis revealing that evening and late-night hours account for the majority of impulse purchases, and food delivery is the dominant regret category.
- NLP analysis of open-ended regret descriptions using TextBlob sentiment scoring and word frequency extraction.
- K-Means clustering identifying three distinct spending behavioral archetypes.
- A Random Forest classifier predicting high financial regret risk with identification of the most predictive behavioral features.

2 Related Work

2.1 Pain of Paying and Payment Abstraction

Prelec and Simester [2] experimentally demonstrated that payment method abstraction reduces the psychological cost of spending. Using credit cards versus cash, they showed that more abstract payment modes increase willingness to pay. UPI extends this abstraction further: no physical medium is involved, and the transaction completes in under three seconds. This theoretical framework directly motivates our hypothesis that UPI use is associated with elevated impulsive purchasing behavior.

2.2 Mobile Payments and Impulse Buying

Research on mobile payment platforms and impulse buying has grown since the proliferation of digital wallets. Studies in the mobile commerce literature identify convenience, reduced transaction time, and cashback rewards as key facilitators of unplanned purchases [3]. In the Indian context, UPI cashback incentives and limited-time offers from apps like Zomato and Blinkit create promotional triggers that encourage spending beyond planned budgets.

2.3 Student Financial Behavior

College students represent a financially vulnerable demographic: income is limited, financial literacy is often low, and social influences

on spending are strong. Prior work has documented that students are susceptible to peer-influenced purchasing (FOMO), emotional spending during academic stress periods, and habitual late-night online consumption facilitated by mobile devices [1].

3 Methodology

3.1 Survey Design

A structured Google Form was designed with six sections:

- **Section A – Demographics:** Age group, gender, college year, monthly budget range, exact monthly budget, income source.
- **Section B – UPI App Usage:** Primary UPI application, perception of UPI as an overspending risk, and reasons for offline UPI use.
- **Section C – Spending Behavior:** Weekly transaction count (range and exact), percentage of unplanned transactions, time-of-day profile for impulse purchases, and balance-checking habits.
- **Section D – Financial Habits:** Expense tracking behavior, frequency of running out of money before month end, and hidden purchase behavior.
- **Section E – Impulse Triggers (Likert Scale):** Eight items rated 1 (Strongly Disagree) to 5 (Strongly Agree): boredom-driven purchasing, FOMO/social influence, late-night scrolling, cashback justification, stress relief buying, scarcity notifications, cart abandonment return, and exam-season stress spending.
- **Section F – Regret Mapping:** Monthly regret frequency, regret category (multi-select), post-purchase regret intensity (1–10 scale), post-regret coping action, and an open-ended description of a regretted purchase.

All humor-labeled answer choices (e.g., “I am scared of what I’ll find”) were preserved to encourage honest responses and later standardized during cleaning.

3.2 Data Collection

The survey was distributed across four college groups between March and April 2026:

- IIT Bhilai (54 responses)
- Jaipur Colleges (30 responses)
- AIIMS Jodhpur (12 responses)
- IIT Delhi (9 responses)

Distribution channels included WhatsApp broadcast, direct sharing in college networks, and the Reddit communities r/IndianTeens and r/personalfinanceindia. A total of **105 valid responses** were collected. Respondent email addresses were collected optionally and stored separately for report delivery, never used in analysis.

3.3 Data Cleaning and Preparation

All preprocessing was performed in Python using the Pandas library. The key steps were:

- (1) **Respondent ID assignment:** Each response was assigned a unique numeric ID with a prefix indicating the source institution.

- (2) **Joke label standardization:** All answer options containing appended humor text were stripped to retain only the categorical value (e.g., “Never (teach us your ways)” → “Never”).
- (3) **Ordinal encoding:** Categorical ordinal columns were mapped to integers: balance_check_habit (Never=0, Rarely=1, Sometimes=2, Always=3), tracks_expenses (No=0, Tries but quits=1, Yes=2), ran_out_of_money (Never=0, Rarely=1, Occasionally=2, Regularly=3), hidden_purchase (No=0, Maybe=1, Yes=2), regret_frequency (Never=0 through 10+=4).
- (4) **Numeric midpoint imputation:** For range-type fields (monthly budget range, weekly transaction range, % unplanned), exact numeric values were used where provided; midpoints of the stated range were substituted otherwise.
- (5) **Multi-select binary encoding:** The impulse_time and regret_categories multi-select fields were exploded into binary flag columns (e.g., flag_latenight, cat_food_delivery).
- (6) **Outlier capping:** Four respondents reported exact monthly budgets exceeding Rs.10,000 within the Rs.6,000+ bracket; these were capped at Rs.7,000.
- (7) **Target variable creation:** high_regret was defined as 1 if regret_intensity ≥ 6, else 0.
- (8) **Impulse composite score:** The mean of all eight Likert trigger columns per respondent, producing a continuous behavioral index.

The final cleaned dataset contains 105 rows and 46 columns with only 23 null values, all in the optional open-ended text column.

3.4 Analytical Methods

3.4.1 Exploratory Data Analysis. Descriptive statistics and visualizations were produced using Matplotlib and Seaborn. Charts include demographic distributions, spending behavior frequency plots, correlation heatmaps, and gender-disaggregated trigger comparisons.

3.4.2 Natural Language Processing. TextBlob was used to compute polarity ($p \in [-1, 1]$, where -1 is most negative and $+1$ is most positive) and subjectivity ($s \in [0, 1]$, where 1 is maximally subjective) for each open-ended regret description. Word frequency analysis was performed after tokenization and stopword removal using NLTK. A word cloud was generated from the corpus using the wordcloud Python library.

3.4.3 K-Means Clustering. K-Means partitions n observations into k clusters by minimizing within-cluster sum of squares (WCSS):

$$\arg \min_S \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \quad (1)$$

where S_i is the set of points in cluster i and $\boldsymbol{\mu}_i$ is the cluster centroid. Features were standardized using Z-score normalization prior to clustering:

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

The optimal number of clusters k was determined using the elbow method, plotting WCSS against k from 2 to 11 and identifying the inflection point. Principal Component Analysis (PCA)

was applied to reduce the feature space to two dimensions for visualization.

3.4.4 Random Forest Classifier. A Random Forest is an ensemble of T decision trees, each trained on a bootstrap sample of the data with a random subset of features at each split. The final prediction is obtained by majority vote:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T \quad (3)$$

where h_t is the prediction of the t -th tree. Feature importance is measured by the mean decrease in Gini impurity across all trees:

$$\text{Importance}(f) = \frac{1}{T} \sum_{t=1}^T \sum_{n \in N_t(f)} \Delta G(n) \quad (4)$$

where $N_t(f)$ is the set of nodes in tree t that split on feature f and $\Delta G(n)$ is the reduction in Gini impurity at node n . The model was trained with $T = 100$ trees on an 80/20 stratified train-test split.

4 Results

4.1 Demographic Profile

The sample comprised 105 respondents: 71 male (67.6%) and 34 female (32.4%). The majority (91, or 86.7%) were undergraduates; 14 (13.3%) were postgraduates. Age distribution was predominantly 18–21 years (78 respondents, 74.3%). All respondents were college students at Indian higher education institutions.

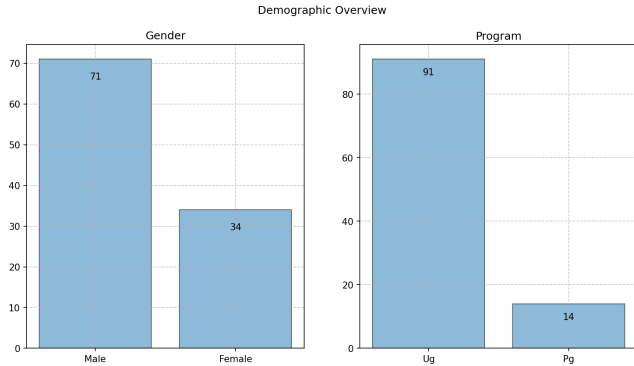


Figure 1: Demographic distribution: Gender (left) and Program (right).

The primary UPI application was Google Pay (GPay) used by 44.8% of respondents, followed by PhonePe (26.7%) and Paytm (14.3%). The three minor-use apps (Bhim UPI, Navi UPI, Other) each constituted 4.8%.

4.2 Spending Behavior

The largest budget segment was Rs.1,000–3,000 per month (50 respondents, 47.6%), consistent with a predominantly parent-funded student population.

Transaction frequency was highest in the 4–7 per week bracket (40 respondents, 38.1%), with 19% making 15 or more transactions weekly. Regarding unplanned purchases, 37.1% reported 20–40% unplanned and 39 respondents (37.1%) reported 40–60% unplanned,

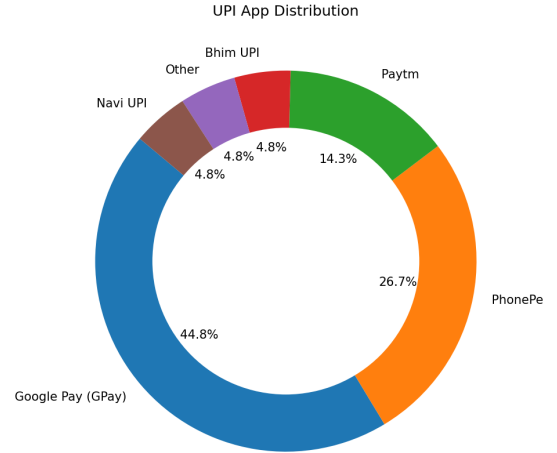


Figure 2: UPI application market share among respondents.

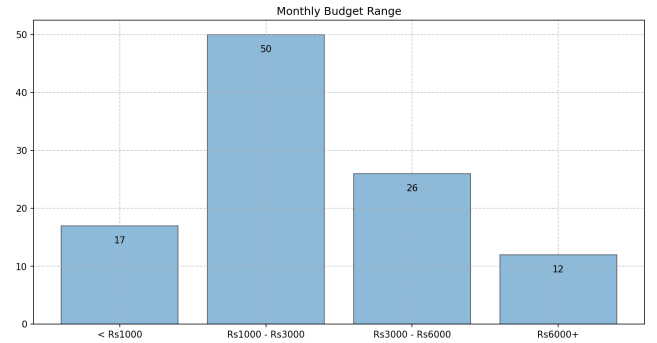


Figure 3: Monthly budget range distribution across respondents.

indicating that the modal respondent makes roughly half their transactions without prior intent.

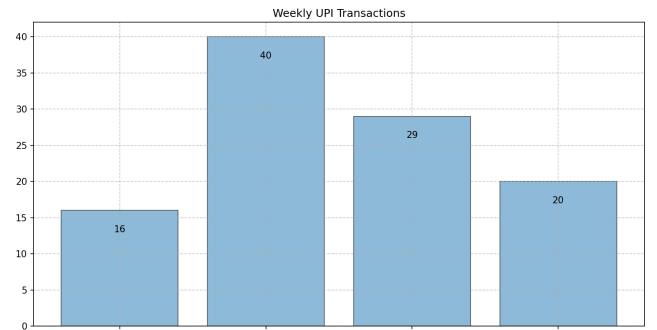


Figure 4: Weekly UPI transaction frequency distribution.

The timing of impulse purchases is strongly concentrated in the evening and late-night hours, as shown in Figure 5. Of 105 respondents, 73 (69.5%) reported making impulse purchases in the evening, and 42 (40%) during late night (10 PM–2 AM). Post-midnight purchases were reported by 9 respondents (8.6%).

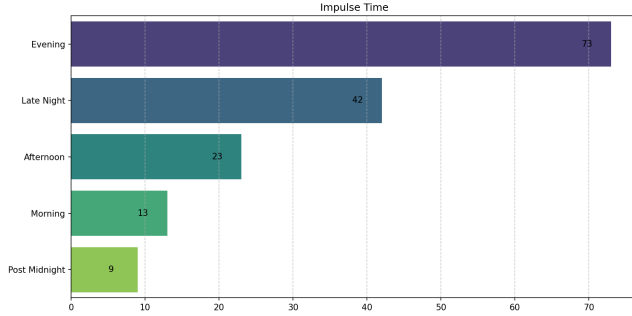


Figure 5: Time-of-day profile for impulse purchases (multi-select).

4.3 Regret Patterns

Monthly regret frequency was low for most respondents, with 51 (48.6%) reporting 1–2 regretted purchases per month and 28 (26.7%) reporting none. However, 26 respondents (24.8%) reported 3 or more regretted purchases monthly.

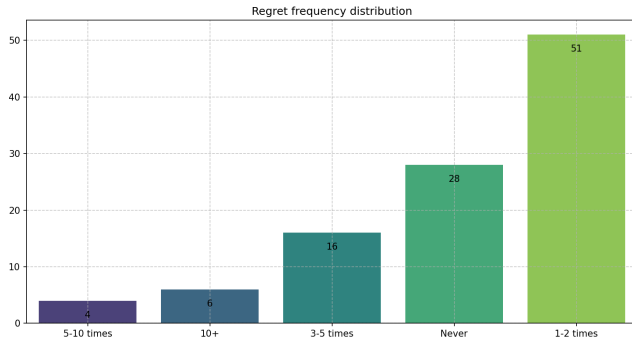


Figure 6: Monthly regret frequency distribution.

Food delivery (Swiggy/Zomato) was the most regretted category (51 respondents), closely followed by offline cafe and canteen spending (46). Grocery quick-commerce (Blinkit/Instamart) ranked third (33).

Regret intensity, rated 1–10, had a mean of 4.35 (SD $\approx$ 2.6). The distribution shows a concentration at intensity 3 (19 respondents) and a secondary cluster at intensity 7 (11 respondents). Using a threshold of 5.5, respondents were classified as high regret ($\geq$ 5, $n = 35$, 33.3%) or low regret ($\leq$ 5, $n = 70$, 66.7%).

4.4 Impulse Trigger Analysis

All eight Likert trigger items were scored 1–5. Mean scores are reported in Table 1.

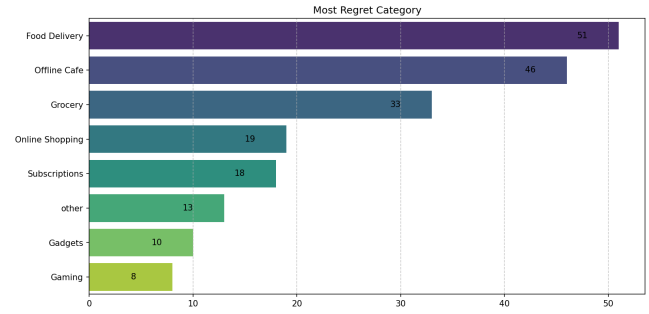


Figure 7: Regret category distribution (multi-select, sorted by frequency).

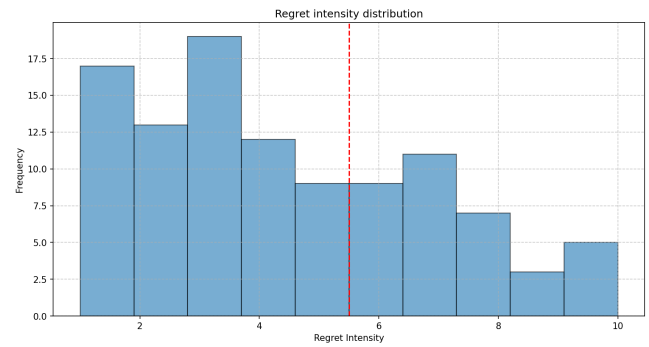


Figure 8: Regret intensity distribution (1–10) with high-regret threshold at 5.5.

Table 1: Mean Likert Scores for Impulse Trigger Items (1–5 scale)

Trigger	Mean Score	Max Observed
Stress Relief Buying	2.07	5
Exam Season Spending	2.06	5
Cashback Justification	1.91	5
FOMO / Social Influence	1.71	5
Late-Night Scrolling	1.70	5
Boredom-Driven Buying	1.57	5
Scarcity Notifications	1.53	4
Cart Abandonment Return	1.41	4

Stress relief and exam season spending ranked highest, indicating that academic pressure is a more significant impulse trigger than social influence or promotional mechanics for this sample. A gender-disaggregated comparison reveals that females score slightly higher on stress relief buying while males score higher on exam season spending.

The mean impulse composite score (mean of all 8 triggers) across the full sample was 1.74 out of 5, indicating that on average respondents leaned toward disagreement with impulsive trigger statements. However, the distribution was right-skewed, meaning a minority of students showed substantially elevated composite scores.

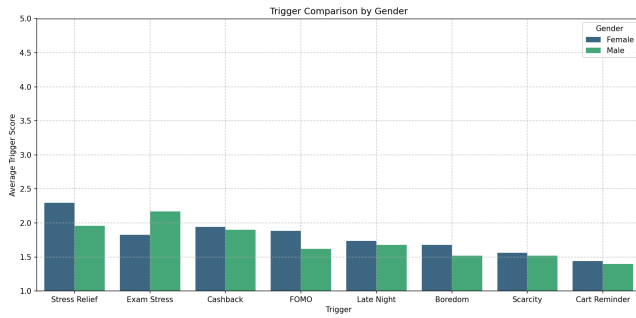


Figure 9: Average impulse trigger scores by gender (scale 1–5, y-axis starts at 1.0).

The correlation heatmap (Figure 10) reveals the strongest positive correlation between regret frequency and regret intensity ($r = 0.38$), and between regret frequency and running out of money ($r = 0.33$). Balance checking habit shows a small negative correlation with unplanned purchasing ($r = -0.15$), consistent with the expectation that financial mindfulness reduces impulsive spending.

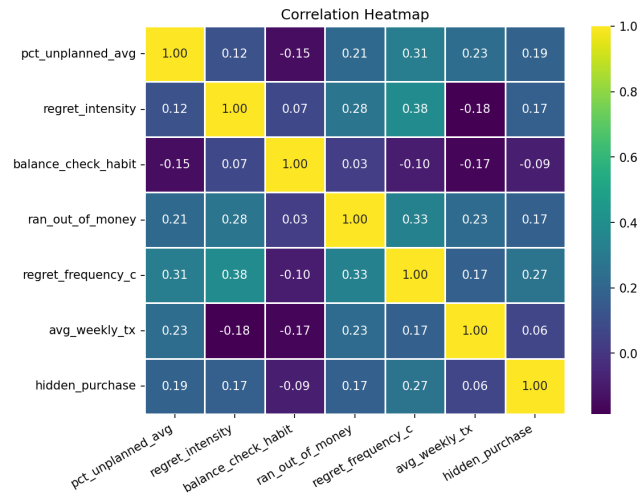


Figure 10: Pearson correlation heatmap of key numeric behavioral variables.

4.5 Hidden Purchase Behavior

42% of male respondents and 35% of female respondents reported having hidden a purchase from parents or roommates. This behavioral indicator captures implicit guilt associated with a purchase and was included as a feature in the predictive model.

4.6 NLP Analysis

Of 105 respondents, 21 provided a response to the open-ended question: "Describe a recent online/UPI purchase you regretted. What did you buy? What were you feeling right before you bought it? What made you click Pay?"

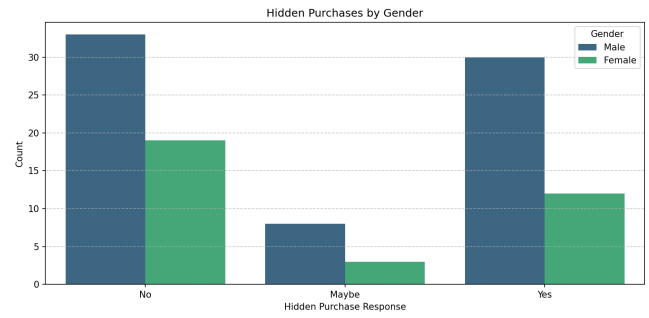


Figure 11: Hidden purchase behavior by gender (No, Maybe, Yes).

TextBlob sentiment analysis yielded a mean polarity of 0.025 (near neutral with slight positive skew) and mean subjectivity of 0.333 (low-to-moderate subjectivity). The Pearson correlation between polarity score and self-reported regret intensity was $r = 0.020$, indicating that the linguistic sentiment of regret descriptions did not strongly predict quantitative regret intensity in this small sample.

Due to the limited corpus size, these sentiment observations should be interpreted qualitatively rather than as statistically robust linguistic findings.

Word frequency analysis (Figure 12) of the cleaned, stopword-removed corpus identified *food* as the highest-frequency term (9 occurrences), followed by *bought*, *feeling*, *money*, and *didn't*. The presence of *return*, *regret*, and *dont* confirms that descriptions were genuinely regret-oriented.

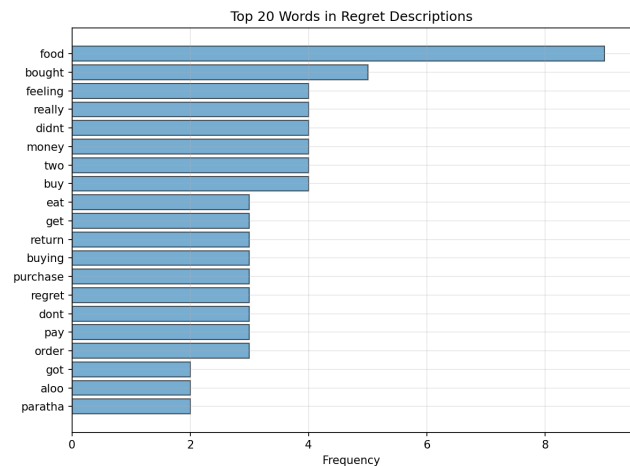


Figure 12: Top 20 words in open-ended regret descriptions after stopword removal.

Manual qualitative tagging of the 21 descriptions into trigger categories (Figure 14) identified that Emotional (19%), Poor Planning (19%), Cravings (19%), and Need (19%) were the four most common triggers, collectively accounting for 76% of descriptions. Social

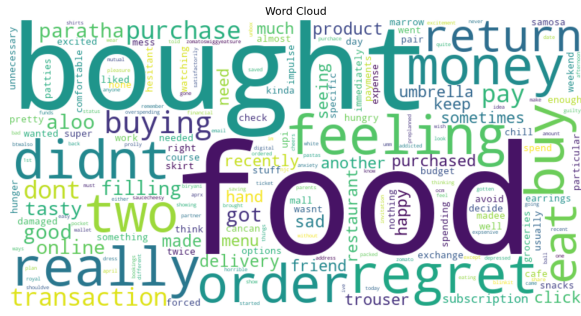


Figure 13: Word cloud generated from all 21 open-ended regret descriptions.

Influence (9.5%) and promotional triggers such as Offers (4.8%) were less prominent.

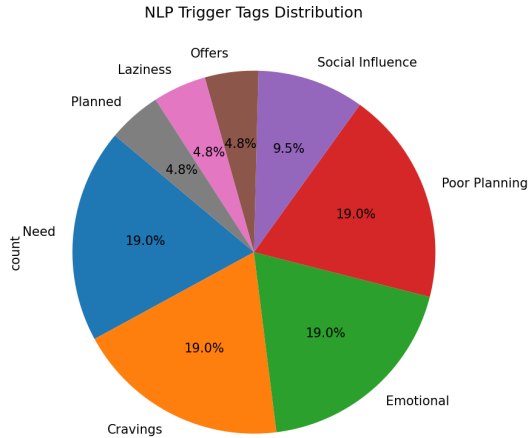


Figure 14: Qualitative trigger tag distribution across 21 open-ended regret descriptions.

4.7 K-Means Clustering

The elbow method (Figure 15) identified $k = 3$ as the optimal number of clusters, evidenced by the inflection in the WCSS curve between $k = 2$ and $k = 4$ with diminishing returns thereafter. The feature matrix included 13 standardized variables: impulse composite score, percent unplanned, average weekly transactions, regret frequency, balance check habit, ran out of money, hidden purchase, tracks expenses, and five time-of-day flags.

The three clusters and their centroid profiles are summarized in Table 2.

Routine Evening Spenders (62.9%): Moderate impulsivity, low late-night activity, spend primarily in the evening. Regret frequency is low. This is the normative college student profile.

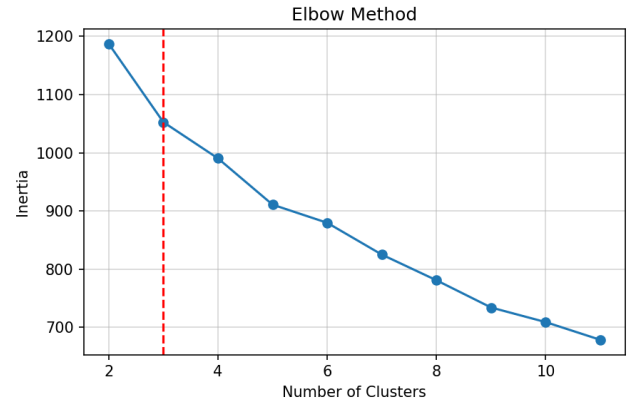


Figure 15: Elbow method for K-Means: WCSS vs. number of clusters. Optimal $k = 3$ marked.

Table 2: K-Means Cluster Centroids (Key Features)

Feature	Routine (n=66)	Controlled (n=31)	High Impulsive (n=8)
Impulse Composite	1.59	1.93	2.25
% Unplanned	35.5	33.9	65.0
Avg Weekly Tx	10.0	9.2	15.8
Regret Frequency	1.09	1.00	2.00
Ran Out of Money	1.17	1.52	2.75
Hidden Purchase	0.97	0.55	1.75
Flag: Late Night	0.24	0.58	1.00
Flag: Post Midnight	0.02	0.00	1.00

Controlled Spenders (29.5%): Lower hidden purchase behavior and slightly higher balance-checking discipline. Late-night purchasing is moderate. These respondents show greater financial awareness.

High Impulsive Spenders (7.6%): Substantially elevated on all risk indicators. All eight respondents in this cluster reported late-night purchasing and post-midnight purchasing, 65% unplanned transactions, the highest ran-out-of-money frequency (mean 2.75/3), and the highest hidden purchase rate (1.75/2). This cluster exhibits the strongest behavioral indicators associated with impulsive digital spending and financial regret.

4.8 Random Forest Classification

A Random Forest classifier ($T = 100$, $\text{random_state} = 42$) was trained to predict `high_regret` (binary) on an 80/20 stratified split (84 train, 21 test). Results are reported in Table 3.

Overall accuracy was 62%. The lower F1 for the high regret class (0.33) reflects class imbalance (70 low vs. 35 high) and the small test set size. The model demonstrates stronger predictive performance for low-regret respondents, likely influenced by class imbalance and limited sample size.

Feature importance analysis (Figure 17) reveals that the **impulse composite score** is the single most predictive feature (importance

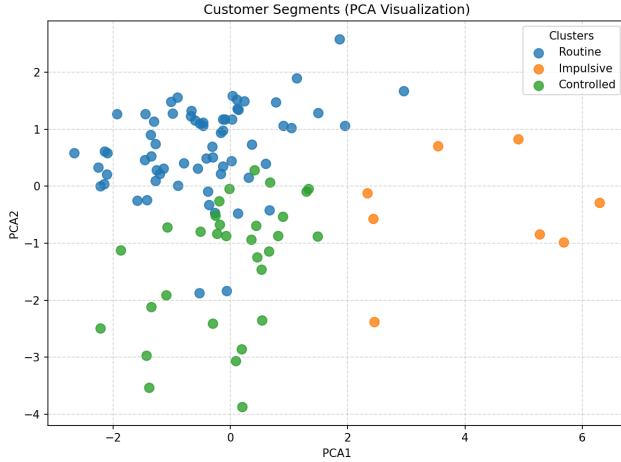


Figure 16: PCA visualization of K-Means clusters. High Impulsive Spenders (orange) are clearly separated along PCA1.

Table 3: Random Forest Classification Report

Class	Precision	Recall	F1	Support
Low Regret (0)	0.69	0.79	0.73	14
High Regret (1)	0.40	0.29	0.33	7
Weighted Avg	0.59	0.62	0.60	21

≈ 0.183), followed by **average weekly transactions** (≈ 0.158), **regret frequency** (≈ 0.133), and **running out of money** (≈ 0.118). Time-of-day flags contributed relatively less, though flag_latenight and flag_postmidnight were present in the High Impulsive cluster.

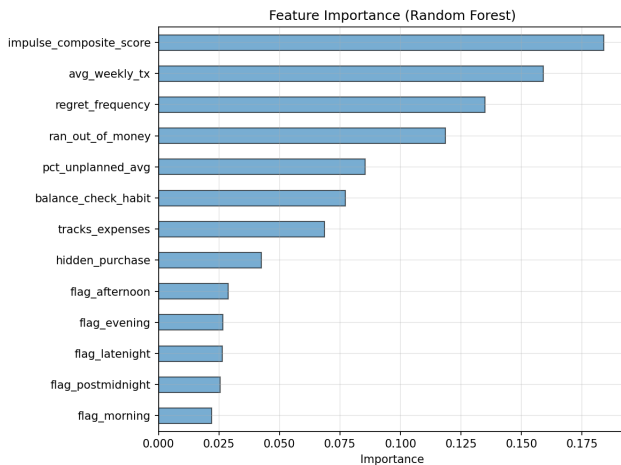


Figure 17: Random Forest feature importance scores, sorted descending.

5 Discussion

5.1 Evening and Late-Night Spending as the Primary Risk Window

The finding that 69.5% of respondents impulse-buy in the evening and 40% during late night is consistent with existing research on digital device usage patterns among college students. The absence of social oversight, combined with access to food delivery and e-commerce platforms that operate 24 hours, creates a structural environment for unplanned spending. The High Impulsive cluster shows 100% late-night and post-midnight purchasing, suggesting that nocturnal UPI activity is a reliable behavioral signature of elevated regret risk.

5.2 Stress and Academic Pressure as Dominant Triggers

Contrary to common assumptions that impulse buying is primarily driven by social influence (FOMO) or promotional mechanics (cash-back, flash sales), the highest-scoring triggers were stress relief buying (mean 2.07) and exam-season spending (mean 2.06). This positions emotional regulation as the central mechanism of UPI impulse behavior in the student context. Students under academic pressure appear to use consumption as a coping mechanism, an observation supported by the NLP analysis where “sad,” “stressed,” and “anxiety” appeared in regret descriptions.

5.3 Food as the Dominant Impulse and Regret Category

Food delivery dominated both the word frequency analysis (“food,” “order,” “eat,” “hungry,” “aloo,” “paratha” in the top words) and the regret category rankings. This is structurally explained by the near-instant delivery windows of Swiggy, Zomato, and Blinkit: the time between the impulse trigger and gratification is 20–40 minutes, providing no natural delay that might allow deliberate reconsideration.

5.4 Feature Importance and Practical Implications

The Random Forest identifies the impulse composite score (a behavioral trait, not a demographic) and transaction frequency (a usage behavior) as the two strongest predictors of financial regret. This implies that interventions should target behavioral patterns rather than demographics. Fintech applications could implement spending nudges triggered by composite trigger score thresholds or by detecting elevated transaction frequency relative to the student’s historical baseline.

6 Limitations

Despite providing meaningful behavioral insights, this study contains several limitations that should be acknowledged.

First, the dataset is based on a relatively small and non-random sample of college students collected primarily from selected Indian institutions. As a result, the findings should be interpreted as exploratory behavioral indicators rather than generalized conclusions applicable to all students or digital payment users.

Second, the survey data is self-reported in nature, which introduces the possibility of response bias and behavioral inconsistency. Some respondents reported relatively low impulsive spending scores while simultaneously exhibiting high levels of financial regret. This suggests that self-perception of impulsive behavior may differ from actual spending outcomes, particularly in emotionally driven or convenience-based purchases.

Third, the demographic distribution of the dataset is imbalanced, with a larger proportion of undergraduate and male respondents. This may influence observed behavioral patterns and limit representational diversity.

Additionally, the NLP component is constrained by the limited number of free-text regret responses available in the survey. Consequently, the natural language analysis should be interpreted as supportive exploratory evidence rather than comprehensive linguistic modeling.

Finally, the machine learning models developed in this project are exploratory and designed for behavioral pattern analysis rather than real-world financial risk prediction. The generated risk scores represent similarity to high-regret behavioral patterns observed in the collected sample and should not be interpreted as actual financial or psychological assessments.

Despite these limitations, the project demonstrates how behavioral analytics, machine learning, and NLP techniques can be integrated to study impulsive digital spending behavior in emerging fintech ecosystems.

The dataset also reflects a relatively low average impulse composite score despite observable regret patterns, suggesting possible behavioral underreporting or mismatch between perceived and actual impulsive spending behavior.

7 Conclusion

This study examined UPI-based impulse spending behavior and financial regret among 105 Indian college students using a multi-method approach including EDA, NLP, K-Means clustering, and Random Forest classification. Key findings are:

- (1) Evening and late-night hours are the primary impulse purchase windows, with 69.5% of respondents reporting evening impulse spending.
- (2) Food delivery is the most common and most regretted spending category.
- (3) Stress relief and academic stress are the dominant behavioral triggers, ranking above social influence and promotional mechanics.
- (4) Three spending archetypes were identified: Routine Evening Spenders (62.9%), Controlled Spenders (29.5%), and High Impulsive Spenders (7.6%), with the latter showing 100% late-night activity and the highest financial distress indicators.
- (5) The impulse composite score and weekly transaction frequency are the strongest predictors of high financial regret risk.

Future work should expand the sample across more institutions and geographic regions, incorporate longitudinal tracking of spending behavior, and explore the effectiveness of digital nudge interventions targeting the identified behavioral signatures.

The complete project repository, notebooks, processed datasets, and dashboard source code are available at:

<https://github.com/rengoku26r/upi-impulse-analysis>

The deployed interactive Streamlit dashboard can be accessed at:

<https://upi-impulse-analysis-263.streamlit.app/>

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